

Weekly Report #2-3

Benchmarking Stream Clustering Algorithms in River

Contributor	Contributor	Reporting period	08 Jun 2026 – 21 Jun 2026
Mentors	Advisors / Mentors	Phase	Phase 1 & 2 - Pipeline, Metrics & Algorithm Integration
Repository	github.com/user/repo	Report date	September 11, 2026

1. Executive Summary

Objective - Set up River and familiarize with River, implement core stream clustering evaluation metrics, implement foundational algorithms, and formulate an objective hyperparameter benchmarking methodology.

Status - DONE Ahead of Schedule

Progress Tracker

Table 1: Comparison between Proposal Schedule and Actual Progress (Weeks 2 & 3).

Phase / Week	Proposal Milestone	Actual Reality (Weeks 2-3)	Status	Completion
P1.2 (Week 2)	Study River algorithms & setup dev env	Installed River, explored framework & pipeline setup	DONE	100%
P2.1 (Week 3)	Establish core benchmark metrics	Implemented 9 metrics (ARI, NMI, AMI, Purity, RI, Silhouette, SSQ, DB, CH)	DONE	100%
P3.1 (Weeks 6-7)	Implement 1st algorithm group (early)	Implemented 10 algorithms (5 foundational + 5 exploratory)	DONE	100%
Strategic Plan	Roadmap alignment	Prioritized foundational algorithms with sound theory/experiments	DONE	100%
Methodology	Benchmark design	Researched LODO framework [1] & proposed Relative Parameter LODO [2]	DONE	100%

Takeaway. Core pipeline setup and evaluation metrics are complete. Algorithm implementation and benchmark methodology formulation are substantially ahead of the original proposal schedule.

Overall: Weeks 2-3/18 - ahead of schedule.

2. Completed Tasks

C1. River Pipeline Setup and Evaluation Metrics Integration (Phases 1 & 2)

- Installed River and established a unified pipeline execution framework to streamline streaming model execution, evaluation, and algorithm integration.
- Implemented and wrapped comprehensive clustering evaluation metrics directly compatible with River streams:
 - **External metrics:** Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), Adjusted Mutual Information (AMI), Purity, Rand Index (RI).
 - **Internal metrics:** Silhouette Coefficient, Sum of Squared Errors (SSQ/SSE), Davies–Bouldin (DB) Index, Calinski–Harabasz (CH) Index.

C2. Stream Clustering Algorithm Implementations and Prioritization Strategy (Phase 3)

- Implemented 10 streaming clustering algorithms:
 - **Foundational, standard, and widely cited:** DStream, HPStream, Birch, RepStream, SVstream.
 - **Exploratory/other models:** DDStream, MrStream, DCUstream, improved_Birch, SOSstream.
- **Implementation Strategy:** Prioritize foundational algorithms first because they possess rigorous theoretical foundations and solid, reproducible experimental setups in original papers, making them ideal for verification. Conversely, other algorithms often present ambiguous algorithmic formulations and questionable experimental reproducibility.

C3. Research on Hyperparameter Benchmarking for Stream Clustering (Methodology)

- **Current State & Challenges:**
 - *Overfitting:* Most literature reports hyperparameters manually tuned specifically on the authors' chosen datasets. On different benchmarks, performance drops drastically or exhibits unexpected runtime/RAM spikes.
 - *Unfair Comparison:* Cerqueira et al. [1] emphasize that heuristic parameter choices or tuning on test data creates severe evaluation bias.
 - *Core Objective:* Identify hyperparameter configurations that perform robustly and objectively across diverse datasets, rather than finding an isolated "best" parameter for a single dataset.
- **Reference Baseline - Leave-One-Dataset-Out (LODO):**
 - *Mechanism (Cerqueira et al. [1]):* Use $n - 1$ datasets to search for optimal configurations (via random search) and evaluate on the remaining held-out dataset. Keeps tuning independent from test data.
 - *Advantages:* Model-agnostic, evaluates wide parameter spaces, straightforward, reproducible.
 - *Drawbacks:* High computational cost, enormous search space, reliance on chance rather than domain insight.
 - *Fatal Limitation for Clustering:* Standard LODO fails because static absolute parameters (e.g., fixed ϵ) lack consistent geometric meaning across datasets with varying dimensionality, scales, and cluster counts (e.g., KDD99 with 23 clusters vs. Shuttle with 2 clusters). No single static absolute value works universally.
- **Proposed Solution - Purposeful LODO with Relative Parameters:**
 - (a) **Tune Key Parameters Only:** Focus exclusively on 1–2 pivotal parameters (or more depending on algorithm) cluster structure and granularity, for examples:
 - *Centroid (KMeans, etc.):* k (number of clusters).

- *Density (DenStream, etc.)*: ϵ , `clustering_threshold` (micro-cluster radius and merge/split criterion).
 - *Grid (DStream, etc.)*: `grid_width` (cell density threshold).
 - *Hierarchical (BIRCH, etc.)*: `threshold` (CF-tree branching threshold).
 - *Graph (RepStream, etc.)*: α (representative point density).
- (b) **Use Relative Parameters via Data Statistics (`param = f(stats)`)**: Tune relative scaling coefficients instead of static values:
- $\epsilon = c \cdot d$, where d is median pairwise distance or average k -NN distance.
 - `grid_width` = $\text{range}(X)/g$, where g is target divisions per dimension.
 - `threshold` = $c \cdot \sigma$, where σ is mean standard deviation across dimensions.
 - k and α tied to class counts or sample mode estimates.
- (c) **Core Benefit**: Supported by the precedent of Fan et al. [2], a single scaling coefficient c automatically adapts to data density (larger ϵ for sparse data, smaller for dense data), eliminating the flaw of enforcing rigid absolute values across heterogeneous datasets.

3. Results & Visualization

[Section reserved for upcoming experimental results and benchmark visualizations]

4. Issues & Blockers

[No major blockers encountered during Weeks 2–3]

5. Next Week's Plan

[Reserved for Next Week's milestones]

6. References & Links

Code produced in Weeks 2–3

Ref	Description	Link
Module 1	Pipeline setup and Evaluation Metrics (ARI, NMI, Silhouette, etc.)	metrics/river_eval
Module 2	Foundational & Experimental Clustering Algorithms	algorithms/stream_models
Module 3	Hyperparameter Benchmarking Harness (Relative LODO)	benchmarks/relative_lodo

Literature and Documentation

References

- [1] V. Cerqueira, H. M. Gomes, M. Heyden, B. Pfahringer, and A. Bifet, “A Framework for Evaluating and Benchmarking Concept Drift Detection Methods,” in *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26)*, 2026. arXiv:2606.07789.
- [2] X. Fan, Y. Yue, P. Sarkar, and Y. X. R. Wang, “On Hyperparameter Tuning in General Clustering Problems,” in *Proceedings of the 37th International Conference on Machine Learning (ICML 2020)*, PMLR 119, pp. 2965–2975, 2020. URL: [proceedings.mlr.press/v119/fan20b.html](#).
- [3] C.-D. Wang, J.-H. Lai, D. Huang, and W.-S. Zheng, “SVStream: A Support Vector-Based Algorithm for Clustering Data Streams,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 25, no. 6, pp. 1410–1424, 2013. DOI: [10.1109/TKDE.2011.263](#).

7. Reproducing the SVStream Experiments

We reproduced the experimental study of SVStream [3] (Figures 3–18) with our River implementation and benchmark framework. This section summarises the setup, where the results agree with the paper, where they do not, and why.

7.1 Setup

Table 2: Datasets: paper vs. reproduction.

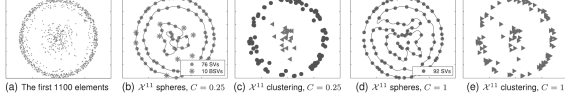
Dataset	Paper	Reproduction
Ring-Ball	2,000 points, 2 overlapping classes	Re-created (data not released): ring $R = 3$, ball $\sigma = 0.39$ with a 5% wider halo. Coordinate scale unknown.
Smileface-Twomoons	7,000 points, 5 classes, 3.4% noise	Re-created with the same phases: moons appear at chunk 12, eyes converge from chunk 51.
KDD CUP'99	494,020 records, 34 attributes	First 10,000 records, online z-score.
Forest CoverType	581,012 records, 10 attributes	First 10,000 records, online z-score.

Each chunk of $M = 100$ elements is learned first and then labelled; quality is the Rand index (RI) of each chunk, noise excluded, averaged over the stream. We use $\delta = 0.6$, $\eta = 1$, $\zeta = 2$ and the (q, C) of Table 1 on the synthetic streams. On the standardised real streams the paper's q (0.011 and 16) makes the kernel vanish, so $q = 0.02$ (KDD) and $q = 0.15$ (CoverType) are used instead.

7.2 Main Results

Table 3: Paper findings vs. our measurements for every figure.

Fig.	Paper	Reproduction	Match
3	Two synthetic streams	Re-created, visually similar	Input
4	$C = 0.25$: 76 SVs, 10 BSVs; $C = 1$: one cluster	$C = 0.25$: 178 SVs, 0 BSVs ; $C = 1$: one cluster	Partly
5–6	Avg. RI 0.9267	Avg. RI 0.9179	Yes
7–12	Avg. RI 0.9857; 53–126 SVs+BSVs	Avg. RI 0.7624 ; 66→246 SVs, 0 BSVs; collapse after chunk 51	No
13	Detection 82–98%, false alarms < 6%	0% at $C = 0.3$; at $C = 0.015$ same trends, 5–46% false alarms	Trends
14	RI stable in M ; time grows with M ; ≤ 200 SVs+BSVs	RI stable; time <i>falls</i> with M ; 162–250 SVs+BSVs	Partly
15	Best $q = 9$ (Ring-Ball), $q = 11$ (Smileface)	Best $q = 9$ on Ring-Ball; Smileface still rising at $q = 25$	Ring-Ball
16	Clear optimum in C	Flat in C	No
17	$(\delta, \eta) = (0.6, 1.0)$ best	η has no effect; $\delta = 0.1$ best	No
18	SVStream best on all streams, smallest memory	Best on Ring-Ball and KDD; behind baselines on Smileface and CoverType	Partly


 Fig. 4. Comparing the performances of SVStream on Ring-Ball with ($C = 0.25$) and without ($C = 1$) BSVs.

(a) Paper

(b) Reproduction

Figure 1: Figure 4, Ring-Ball. $C = 1$ merges the two classes as in the paper, but at the paper's $C = 0.25$ no BSV is ever created.

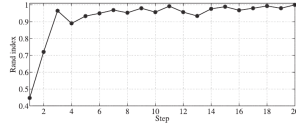
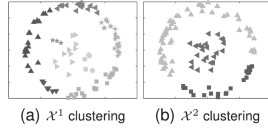


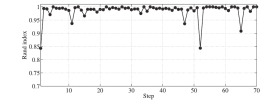
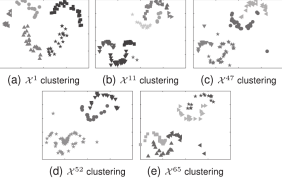
Fig. 5. The Rand index as a function of the chunk step by SVStream on the Ring-Ball data stream.


 Fig. 6. The two erroneously clustered chunks on Ring-Ball with $M = 100$, $C = 0.25$ and $q = 9$.

(a) Paper

(b) Reproduction

Figure 2: Figures 5–6, Ring-Ball. Average RI 0.918 vs. 0.927; as in the paper, only the first chunks are poor.


 Fig. 11. The Rand index as a function of the chunk step on Smileface-Twomoons with $M = 100$, $q = 11$, $C = 0.3$, and $\zeta = 2$.

 Fig. 12. All the five mistakenly clustered chunks on Smileface-Twomoons with $M = 100$, $q = 11$, $C = 0.3$, $\zeta = 2$.

have been mistakenly clustered by either merging or splitting some clusters. These five mistakenly classified

(a) Paper

(b) Reproduction

Figure 3: Figures 11–12, Smileface-Twomoons. Near-perfect before the moons appear (RI 0.998), then degrading until everything merges into one cluster in chunks 62–69.

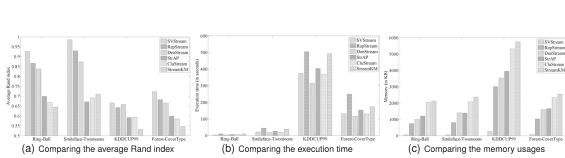


Fig. 18. Comparing the performances of SVStream with five algorithms.

(a) Paper

(b) Reproduction

Figure 4: Figure 18, against River baselines. SVStream clearly wins on Ring-Ball but not on Smileface-Twomoons or CoverType.

7.3 Why the Results Differ

1. **The box constraint of Eq. (5) never binds.** With $\sum_i \beta_i = 1$ and $\beta_i \leq C$, a sphere holds at most $\lfloor 1/C \rfloor$ BSVs (4 at $C = 0.25$), yet the paper shows 10 BSVs at $C = 0.25$ and 14 at

$C = 0.3$, so its C is probably scaled differently (e.g. $1/(NC)$ as in support vector clustering). Implemented literally, β_i stays far below C : no BSVs are created, BSV decay never fires, and noise is kept as SVs. This explains Figures 7–13 and 16: a growing summary, the late collapse on Smileface, 0% outlier detection and flat C curves.

2. **q depends on the unpublished data scale.** The kernel only sees $q\|\mathbf{x}_i - \mathbf{x}_j\|^2$, so rescaling the data by α is the same as multiplying q by α^2 . Our Ring-Ball scale happens to match the paper (best $q = 9$); our Smileface scale does not (best $q > 25$).
3. **Real streams are small, standardised prefixes.** Only 10,000 of about 0.5M records are used. KDD prefix chunks contain one or two classes, which inflates RI (0.982 vs. 0.666 in the paper); on CoverType SVStream effectively forms a single cluster per chunk (RI ≈ 0.33 for every q and C).
4. **Implementation and measurement.** Python/River code and River baselines (StrAP unavailable, DenStream ε rescaled), memory counted on the model object only, and more QP solves for smaller M , which is why runtime falls as M grows.

Takeaway. SVStream reproduces on Ring-Ball (RI 0.918 vs. 0.927, best $q = 9$), but most other results diverge, mainly because the literal C of Eq. (5) never creates BSVs. The C scaling the authors actually used must be clarified before the remaining comparisons are meaningful.